

The Water Cycle

A two-column reference sheet — reading order should run 1 → 2 → 3 → ... → 8

Part One — The Visible Cycle

(1) Evaporation. The sun heats water in oceans, lakes, and rivers, and the liquid turns into invisible water vapor that rises into the air. Most of the water that enters the atmosphere each day evaporates from the surface of the ocean.

(2) Condensation. As the warm vapor rises, it cools. The cooling vapor gathers around tiny particles of dust and forms the visible droplets we see as clouds and fog. This is the reverse of evaporation.

(3) Precipitation. When the droplets in a cloud grow heavy enough, they fall back toward the ground as rain, snow, sleet, or hail. The form depends mostly on the temperature of the air the droplets pass through.

(4) Collection. Fallen water collects in oceans, lakes, rivers, and streams, or it soaks into the soil. From these collecting places the cycle begins again, and the same water is used over and over.

Part Two — The Hidden Cycle

(5) Groundwater. Some precipitation seeps down through the soil and rock until it reaches a layer it cannot pass. This stored underground water, called groundwater, supplies wells and feeds springs during dry seasons.

(6) Transpiration. Plants draw water up through their roots and release it as vapor through tiny openings in their leaves. A single large tree can transpire hundreds of liters of water on a hot day.

(7) Runoff. Water that does not soak in flows downhill across the surface as runoff, carving channels and carrying sediment toward larger bodies of water. Runoff is fastest on steep, bare, or paved ground.

(8) Human Impact. Cities, farms, and dams change how water moves through the cycle. Paving land speeds runoff, irrigation moves water far from its source, and pollution can follow the water through every stage.

Average residence time of water by reservoir

Reservoir	Typical residence time
Atmosphere	About 9 days
Rivers	About 2 weeks
Soil moisture	1 to 2 months
Deep groundwater	Up to 10,000 years